

Class 10

Time dependent simulations II.

Last week: TEBD

- efficient for nearest neighbor interacting 1D chains
 - longer ranged couplings $\Rightarrow$ swap gates
 - $\rightarrow$ More gates (cost?)
 - $\rightarrow$ artificially increased entanglement for swap gates
- $\Uparrow$
accuracy?

This week: "tangent space methods"

Idea: searching $|\psi(t)\rangle$ in the manifold of fixed M MPS states



$$i\partial_t |\tilde{\psi}(t)\rangle = P_{\tilde{\psi}} H |\tilde{\psi}(t)\rangle$$

$\nearrow$
Projects on
the tangent space
at $|\tilde{\psi}(t)\rangle$

The "Basis Update & Galerkin" method

$$\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$$

$$|\psi\rangle = \sum_{i=1}^{\text{dim}_A} \sum_{j=1}^{\text{dim}_B} \psi_{ij} |i\rangle |j\rangle$$

Low rank approximation

$$\psi_{ij} = \sum_{\alpha, \beta=1}^M U_{i\alpha} S_{\alpha\beta} V_{j\beta}$$

$$M \ll \text{dim}_A$$

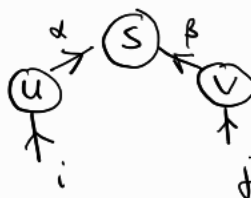
 $M \ll \text{dim}_B$

U and V are isometries:

$$\sum_i U_{i\alpha} U_{i\tilde{\alpha}}^* = \delta_{\alpha\tilde{\alpha}}$$

$$\sum_j V_{j\beta} V_{j\tilde{\beta}}^* = \delta_{\beta\tilde{\beta}}$$

$$\leadsto \sum_{\alpha, \beta} |S_{\alpha\beta}|^2 = \langle \psi | \psi \rangle = 1$$

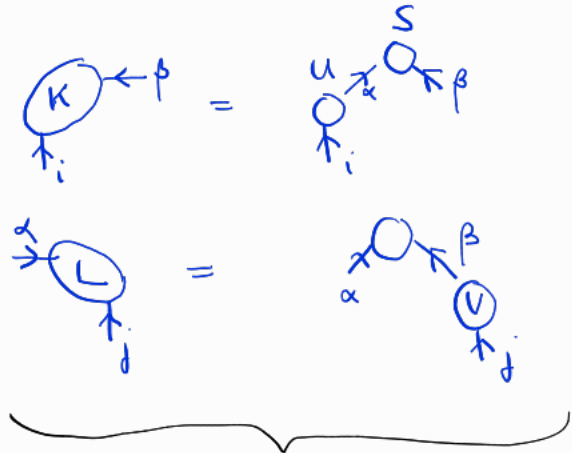


$\Rightarrow$ Direct evolution of U , S , and V is numerically ill
 $\hookrightarrow$ small Schmidt values of $S \leadsto$ huge curvature of the manifold ($\sim S^{-1}$)
 $\Updownarrow$
 extremely small Δt would be necessary
 $\hookrightarrow$ keeping unitarity of U and $V \iff$ complicated constraint.

$\Rightarrow$ Trick:

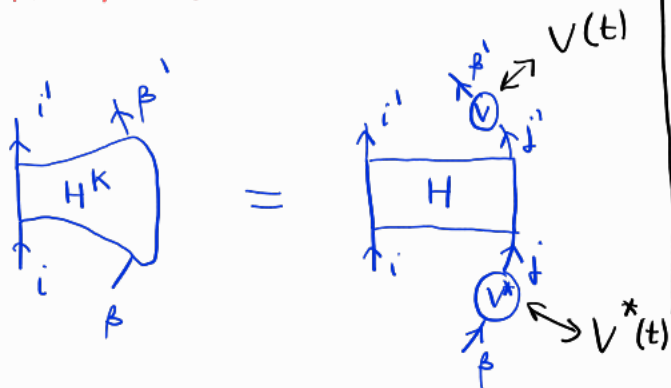
$$K_{i\beta} = \sum_{\alpha} U_{i\alpha} S_{\alpha\beta}$$

$$L_{\alpha j} = \sum_{\beta} S_{\alpha\beta} V_{j\beta}$$



1) K-step:

$$K(t+\Delta t) = e^{-iH^K \Delta t} K(t)$$



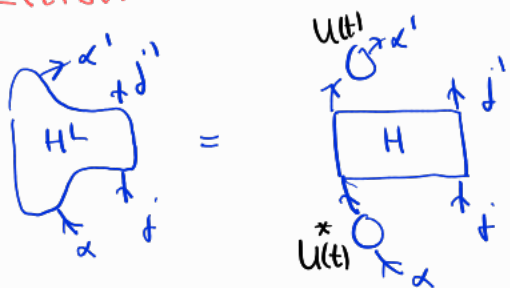
K is the wavefunction projected to $\mathcal{H}_A \otimes \mathcal{H}_B^{trunc}$

L is the wavefunction projected to $\mathcal{H}_A^{trunc} \otimes \mathcal{H}_B$

S is the wavefunction projected to $\mathcal{H}_A^{trunc} \otimes \mathcal{H}_B^{trunc}$

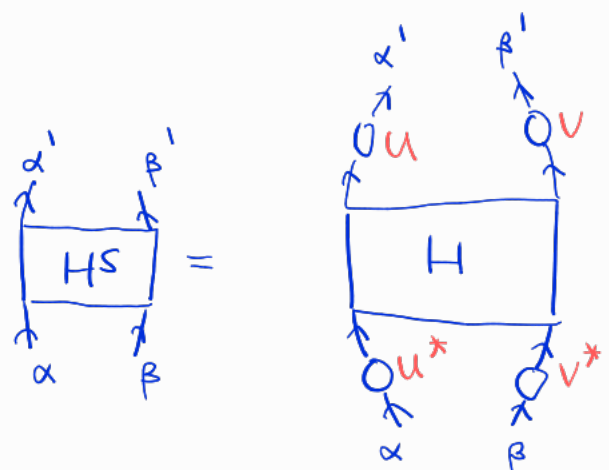
2) L-step:

$$L(t+\Delta t) = e^{-iH^L \Delta t} L(t)$$



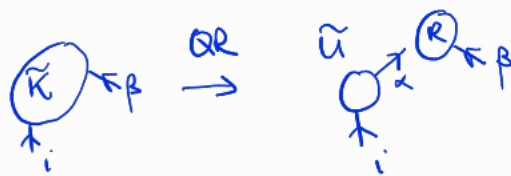
3) S-step

$$S(t+\Delta t) = e^{-iH^S \Delta t} S(t)$$

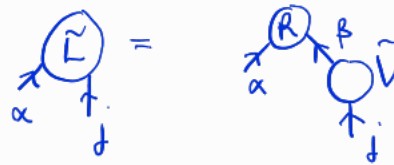


4. new U and V ?

$\tilde{U}(t+\Delta t)$ from QR decomposition of $K(t+\Delta t) = \tilde{K}$



$\tilde{V}(t+\Delta t)$ from "RQ" decomposition of $L(t+\Delta t) = \tilde{L}$



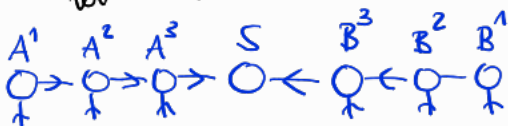
Variants: what U and V is used in the S-step?

- "explicit Euler": $\bar{U} = U(t)$ $\bar{V} = V(t)$ we use the "old"
- "leapfrog": $\bar{U} = \tilde{U}(t+\Delta t)$ $\bar{V} = \tilde{V}(t+\Delta t)$ we use the "new"

$\hookrightarrow S(t) \rightarrow \tilde{U}^\dagger U(t) S(t) V(t) \tilde{V}^\dagger$
small initial projection error!

- both: $\bar{U} = \text{orth}[U(t), \tilde{U}]$
 $\bar{V} = \text{orth}[V(t), \tilde{V}]$ }
 - Bond dimension is doubled
 - both "old" and "new" states are used
 - Adaptive increase of M is possible.
- Truncation after S-step was performed

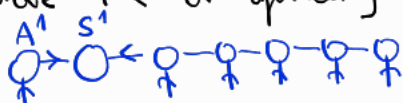
$\rightarrow$ BUG for MPS



K-step and L-step is modified: iterative updates for A 's and B 's

"K-sweep"

$\hookrightarrow$ move the orthogonality center to left



$\hookrightarrow$ form $K_1 = A^1 S^1$ and perform K-step for K_1

↳ form $\bar{A}^1$ by orthogonalizing the old and new states

↳ move orthogonality center 1 step right

↳ do the same for A_2 and then for $A_3 \dots$

"L-sweep": K-sweep is mirrored

↳ bond dimension is doubled!

↳ S-step is the same as in the 2-site case

↳ truncation after S-step

The Gold-Standard method: TDVP based on projector splitting

TDVP:

$$i\partial_t |\tilde{\Psi}(t)\rangle = \mathcal{P}_{\tilde{\Psi}} H |\tilde{\Psi}(t)\rangle$$

$\tilde{\Psi}$ is an MPS

$$\underline{M}^{\sigma_1} \underline{M}^{\sigma_2} \dots \underline{M}^{\sigma_N}$$

$\mathcal{P}_{\tilde{\Psi}}$ is the tangent space projector.

Statement (not proven here)

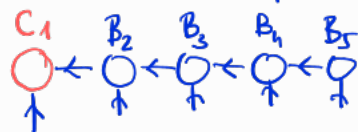
$$\mathcal{P}_{\tilde{\Psi}} = \sum_{n=1}^N \hat{P}_L^{[1..n-1]} \otimes \mathbb{1}_n \otimes \hat{P}_R^{[n+1..N]} - \sum_{n=1}^{N-1} \hat{P}_L^{[1..n]} \otimes \hat{P}_R^{[n+1..N]}$$

↑
↑
↳ "backward step"

Projection on left/right Schmidt states (easy in canonical forms)

↳ Time evolution scheme (1-site TDVP)

i) init MPS: right canonical



evolve $\tilde{C}_1(t+\Delta t) = e^{-iH_1 \Delta t} C_1(t)$

ii) SVD / QR

$$A_1(t+\Delta t) \tilde{R}_1(t+\Delta t) = \text{QR}[\tilde{C}_1(t+\Delta t)]$$

iii) back-step for $\tilde{R}_1$ $R_1(t) = e^{+iH_1 \Delta t} \tilde{R}_1(t+\Delta t)$

iv) $C_2(t) = R_1(t) B_2(t)$

...

$$C_n(t) \rightarrow \tilde{C}_n(t+\delta t) \rightarrow A_n(t+\delta t) \tilde{R}_n(t+\delta t)$$

$$\tilde{R}_n(t+\delta t) \rightarrow R_n(t)$$

$$C_{n+1}(t) = R_n(t) B_{n+1}(t)$$

- IMPORTANT $\leadsto$ there is no back-step at $n=N$!
- After this sweep, we get a fully left-canonical MPS at $t+\delta t$.
- Next sweep: mirrored direction

ISSUE : Bond dimension cannot grow in 1-site TDVP

$\hookrightarrow$ 2-site TDVP (similar to 2-site DMRG)

$L \rightarrow R$

$$\left[\begin{array}{l} C_{n,n+1}(t) \rightarrow \tilde{C}_{n,n+1}(t+\delta t) \rightarrow A_n(t+\delta t) \cdot \tilde{S}(t+\delta t) \tilde{B}_{n+1}(t+\delta t) \\ \tilde{C}_{n+1}(t+\delta t) = \tilde{S}(t+\delta t) \tilde{B}_{n+1}(t+\delta t) \rightarrow \tilde{C}_{n+1}(t) \\ C_{n+1,n+2}(t) = \tilde{C}_{n+1}(t) \cdot B_{n+2}(t) \end{array} \right.$$

$\rightarrow$ extremely similar to DMRG sweep

- instead of diagonalization, there is $e^{-iH_{eff}\delta t}$
- 2-site forward step
- 1-site backward step

Cross-checks for MPS algorithms

- $\rightarrow$ We have devised many algorithms (DMRG, TEBD, BUG, TDVP)
- $\rightarrow$ Complicated tensor-algebra, abstract objects, large gauge freedom

$\Downarrow$

Many error sources.

- $\rightarrow$ How to find errors?
- $\rightarrow$ How to validate results?

Simplest cross-check: **ED** : $\rightarrow$ take a small system ($L \leq 20$)

$\rightarrow$ solve the problem exactly

→ compare it to MPS results

PRO's : • easy and straightforward

• Errors in the MPO Hamiltonian can be found

CON's : • errors caused by inefficient truncation remain usually invisible

FREE-FERMION cross-check

→ specific Hamiltonian:

$$H = \sum_{ij} T_{ij} c_i^\dagger c_j \quad T_{ji} = T_{ij}^*$$

→ specific state (Gaussian)

• Slater determinant (e.g. product state)

• Gaussian density matrix

$$\rho = e^{-\sum_{ij} A_{ij} c_i^\dagger c_j}$$

⇒ Exact results can be found for large systems! ($L \approx 1000$)

Ground state:

$$T_{ij} = \sum_{\alpha} U_{i\alpha} \epsilon_{\alpha} U_{j\alpha}^*$$

$$E_{GS} = \sum_{\alpha=1}^{N_c} \epsilon_{\alpha}$$

Fermi sea

Compare this with DMRG

Time evolution of initial product state

$t=0$ $n_n(t=0) = 0/1$ (bit string)

Solve the 1-particle Schrödinger eq.

$$\hookrightarrow p_i^n(t) = \sum_j [e^{-iTt}]_{ij} p_j^n(0)$$

original position
 $p_j^n(0) = \delta_{jn}$

The many body state is a Slater determinant built from $p_i^n(t)$ 1-particle wavefunctions with $n_n(0) = 1$

$$\langle \psi(t) | n^n | \psi(t) \rangle = \sum_n n_n(t) \cdot |f_n^n(t)|^2$$

We can also calculate von-Neumann entropy (Eisler-Peschel)
ArXiv: 0906.1663

IDEA: Slater determinant is Gaussian.

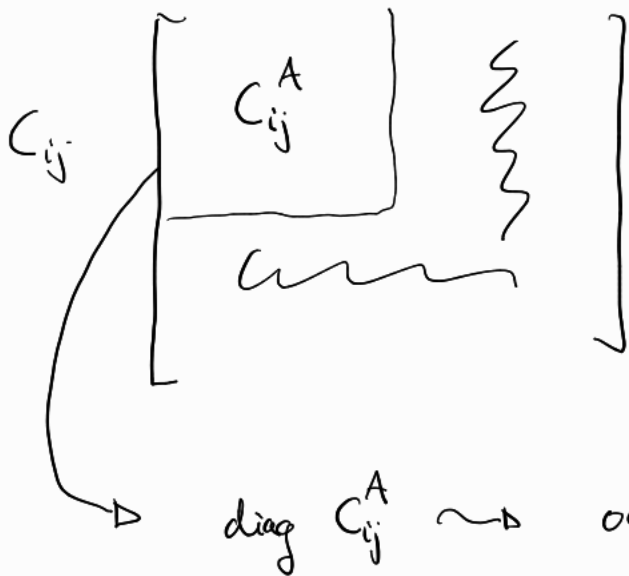
$\Updownarrow$ described by $\langle c_i^\dagger, c_j \rangle$

$$C_{ij}(t) \quad \langle \psi(t) | c_i^\dagger, c_j | \psi(t) \rangle = \langle \psi(0) | \underbrace{U^\dagger c_i^\dagger U}_{\text{backward evolution of ops}} \underbrace{U^\dagger c_j U}_{\text{backward evolution of ops}} | \psi(0) \rangle$$

$$U^\dagger c_i^\dagger U = \sum_j f_j^i(-t) c_j^\dagger$$

IDEA: if the full state is Gaussian, so are the states of sub systems

$$\hookrightarrow \text{Tr}_B |\psi(t)\rangle \langle \psi(t)| = \frac{1}{\mathcal{N}} e^{-\sum_{ij}^A A_{ij} c_i^\dagger c_j}$$



$\Updownarrow$
This is a Gibbs ensemble of free particles

A_{ij} : "entanglement Hamiltonian"

$\text{diag } C_{ij}^A \leadsto$ occupation numbers of block natural orbitals
 $\Downarrow n_\alpha^A$

$$S_{vN} = - \sum_\alpha (n_\alpha \log n_\alpha + (1-n_\alpha) \log (1-n_\alpha))$$